

Sudhendra Kambhamettu

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AWARDS

- Judge at the HackMIT, MIT Boston, Sep 2024
- Judge at the UC Berkeley AI Hackathon Berkeley, Jun 2024
- 2nd Place in MIT EdTech Hackathon Boston, Apr 2024
- 2nd Place - Team Huskies - Reinvent the Wheel Hackathon 2.0 Huntersville, Nov 2022

WORK EXPERIENCE

- Computational Biology (Deep Learning) Co-op, Camp4 Therapeutics, Cambridge, MA** Jul 2023 – Dec 2023
- Engineered AWS framework for LLM-based genomics, boosting ASO up-regulation classification AUPR by 50% and developed custom BPE algorithms to enhance metrics.
 - Led 3-person team in LLM-driven RNA reactivity prediction, achieving 0.17 Average MAE (best 0.13) through fine-tuned architectures for RNA structure analysis.
- Graduate Research Assistant (Multimodality), NULab, Boston, MA** Feb 2023 – Jul 2023
- Developed PyTorch pipeline for multimodal mental health analysis, integrating image and text modalities with fusion architectures, CLIP loss, and SOTA models (BERT, BART, VIT).
 - Pre-trained Vision Transformer on 5M-point multimodal dataset using DINO, achieving ~87% accuracy in attention localization.
- AI Product Developer, RPYD AI GmbH, Hannover, Germany** Dec 2020 – Dec 2021
- Spearheaded the development of the company's key Computer Vision and NLP projects, achieving a no-code AI solution by integrating AWS, GCP, and Azure within the rapyd.ai API framework. Additionally, pioneered a Streamlit application to enhance AI model transparency and user engagement through interactive exploration.
- Computer Vision Intern, RoundSqr, Hyderabad, India** Jun 2020 – Aug 2020
- Coordinated with a specialized team of three in developing an innovative CNN and SVM-based Deep Learning architecture for lipodermatosclerosis detection, achieving ~92% accuracy. This effort notably decreased false positives by 60% in diagnosis procedures, a collaborative achievement with CARE Hospitals, Hyderabad.

PROJECTS

- TARS: Transformative Agent for Recommendation Services** Oct 2024 - Present
- Engineered an AI agent-driven news recommendation system (TARS) utilizing collaborative filtering with GraphDB and content-based filtering with Elasticsearch, enhancing content relevance and user engagement; achieved a 12% increase in NDCG and a 10% rise in MAP using the MIND dataset, while implementing innovative weighting techniques that improved recommendation precision by 15%.
- Evaluating the Proficiency of ChatGPT in Data Structures and Algorithms** Aug 2024 - Oct 2024
- Automated an evaluation pipeline for large language models, assessing proficiency on 569 LeetCode challenges.
 - Optimized GPT-3.5 Turbo with 9 role-based prompts, improving problem-solving efficiency by 20% and achieving an 80% success rate in advanced probability and combinatorics.
- Reinforce Trade Hub: An RL based Equity Portfolio Allocation Tool** Jan 2024 - Mar 2024
- Examining deep reinforcement learning for Equity Quantitative Trading, utilizing Deep Q-Networks (DQN) and Deterministic Policy Gradient (DDPG) techniques to optimize portfolio allocation and enhance investment outcomes.
- LITART: Literature to Artistic Representations with NLP and CV** Feb 2023 - May 2023
- Ideated and created an end-to-end NLP and CV pipeline which extracts named entities from novels, comics etc. and associates their implicit and explicit characteristics together, leveraging them to generate accurate images of fictional characters & scene depiction using a feedback mechanism.

PUBLICATIONS

Kambhamettu, S., Vimal Cruz, M., Anitha, S., Sibi Chakkaravarthy, S. and Nandeesh Kumar, K., 2023 "Brain-Computer Inter-face-Assisted Automated Wheelchair Control Management-Cerebro: A BCI Application." Brain-Computer Interface: Using Deep Learning Applications, pp.205-229 [DOI]

SKILLS

Programming Languages	Python, C++, Java, R, Bash, SQL, BigQuery, Cypher
Frameworks/Tools	PyTorch & Tensorflow end-to-end pipelines, Keras, MySQL, HuggingFace, Langchain, OpenAI, Streamlit, Agents
Cloud Services	AWS, Azure, Docker
Databases	PostgreSQL, Elasticsearch, GraphDB (Neo4j)
Tech Domains	Deep Learning, Computer Vision, Natural Language Processing, Statistical Machine Learning, Multimodality, Reinforcement Learning, Computational Biology, Information Retrieval, RAG

EDUCATION

- M.S. AI, Khoury College of Computer Sciences, Northeastern University** Dec 2024
Algorithms, ML, NLP, HCI, Information retrieval, RL, LLM-based Dialogue Agents
- B.Tech. CS and Engineering, Vellore Institute of Technology (VIT)** May 2022
CGPA: 8.88/10;
Established the first Artificial Intelligence & Robotics Lab in the University.